ALICE L WALTON SCHOOL OF MEDICINE

Heart Blocks: Pathophysiology, Diagnosis and Treatment

Last updated January 13, 2026 · 19 min read · ScholarRx

Adapted by: Diptiman Bose, Phil Riley, Robert Donnell
Editors: Russell Johnson, Jordan Finn

Adapted from: Heart Block and Nodal Disease

Learning Objectives (4) | Versions

After completing this brick, you will be able to:

1. Compare and contrast the clinical presentation and pathophysiology of the different types of heart blocks
2. Differentiate the ECG patterns associated with 1st, 2nd, and 3rd degree heart blocks
3. Differentiate the ECG patterns associated with bundle-branch blocks
4. Outline treat branch blocks.

CASE CONNECTION

JC is a 16-year-old student basketball player who initially presented to an urgent care clinic for a preseason sports physical. Although he had no cardiac symptoms, the urgent care staff ordered an ECG. JC was told it was "not totally normal" and he should see his primary care doctor. JC's vital signs and physical exam are normal. You carefully review the ECG that was done, which shows a regular rhythm, normal axis, good R wave progression, and no ST-T wave changes; his PR interval is slightly prolonged at 220 milliseconds.

What will you tell JC and his parents about his future on the basketball team? Consider your answers as you read, and we'll revisit JC at the end of the brick.

GO TO CONCLUSION ↓

What Is Heart Block?

Heart block is a disorder of the conduction system of the heart in which impulse conduction is blocked or slowed. Although it can occur anywhere in the heart, we are usually concerned with the AV junction (AV block), such that impulses originating from the atria are slowed or interrupted on their way to the ventricles. Depending on where the block occurs, the patient can be asymptomatic and require no treatment, or they might require a pacemaker to prevent profound bradycardia or asystole (cardiac standstill) as a result of the heart block.

Let's start by viewing he heart (Figure 1). It will be helpful to refer to this diagram as you go through the text of this brick.

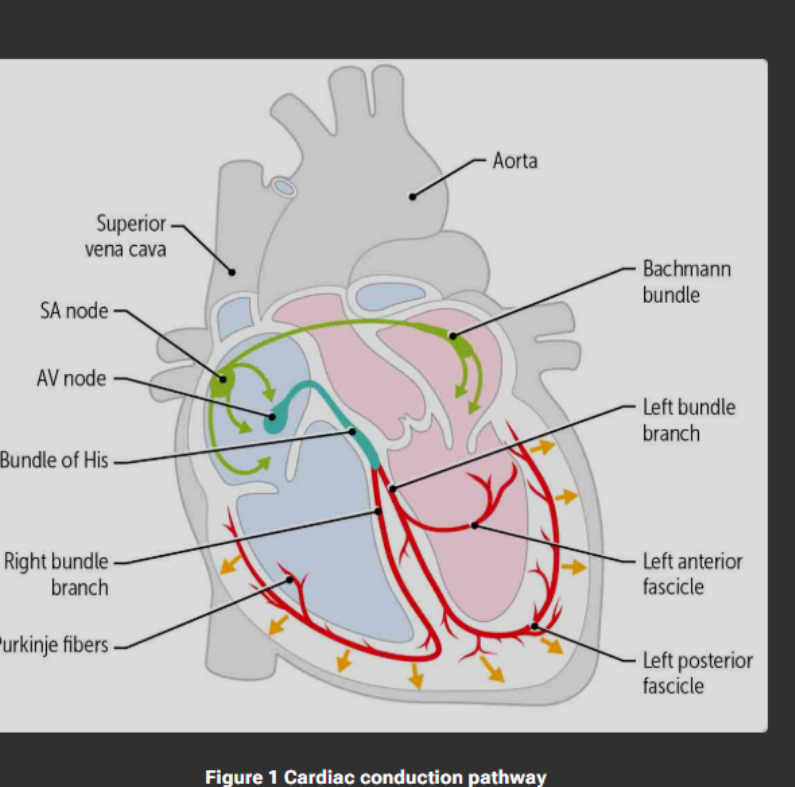


Figure 1 Cardiac conduction pathway

Locations of Heart Block

When we talk about heart block we are usually concerned with AV block, which will be the principal focus of this brick. However, two other forms are worth mentioning: Exit block and bundle branch block.

Exit block is the failure of an te of origin to the surrounding myocardium. Because myocardium is not activated in bulk, blocked impulses are not directly seen on the electrocardiogram (it takes an intracardiac electrode to detect them). Therefore, in clinical medicine utilizing ordinary electrocardiography, the presence of such blocked impulses can only be inferred.

What Is Sinoatrial Node Block (exit block)?

SA block is a form of exit block in which the SA node impulse fails to depolarize the rest of the atrial muscle.

Clinical Presentation

Because it presents on the electrocardiogram as a "sinus pause", it is sometimes lumped in with (and is difficult to distinguish from) sinoatrial node dysfunction which is covered in a separate brick.

Causes. Any pathology in the right atrium can cause SA block.

What is Bundle branch block? (LO#1,3 and 4)

For an understanding of bundle branch block it may be helpful to review previous basic cardiac physiology and anatomy. For this discussion, refer to the figure above. The AV node leads to the Bundle of His. The Bundle of His is sometimes referenced as the "common" Bundle of His, as it branches into the right and left bundle branches. These travel down the right and left sides of the IV septum, respectively. They form extensions of the common Bundle of His. *Anatomically* the nction (because they are actually in the ventricles) but *functionally* they sometimes are, since they participate in transmission of impulses from atria to ventricles.

Conduction velocity varies greatly throughout the AV junction. The first part of the AV junction, the AV node, has the slowest conduction velocity in the entire heart (.05 m/sec). The clinical significance is that the AVN serves as a filter to prevent overly rapid conduction to the ventricles. This serves two important functions. First, it allows a time delay for adequate ventricular filling following atrial contraction (this optimizes cardiac output). Second, should the patient develop atrial flutter or fibrillation, the slow AV nodal conduction limits the number of impulses that can travel to the ventricles, thus avoiding critically fast ventricular rates. (Were it not for this filtering function, atrial flutter and fibrillation would cause ultra fast ventricular rates leading to cardiac arrest).

What is the function of the AV node in cardiac conduction?

Further down in the AV junction, in the Bundle of His and its right and left branches we find the opposite situation: the fastest conduction velocity of the entire myocardium (2-4 m/sec). This ensures that after adequate delay in the AVN, once it is time to activate the ventricles, the activation takes place rapidly and in multiple areas of the myocardium simultaneously. This gives rise to *in parallel* activation of the ventricles and the QRS being sharp and narrow (≤ 100 ms).

So, when conduction in one of the bundle branches is slowed or lost entirely, the QRS becomes abnormally wide and assumes a different morphology. Right and left bundle branch blocks differ in appearance and can be distinguished electrocardiographically.

Right bundle branch block (RBBB)

In RBBB the delay in conduction through the right bundle branch causes depolarization of the heart in *series*, left ventricle followed by right, resulting in a **widening of the QRS complex** (>120 ms in adults). The other characteristic feature is rsR' (R S R prime), resulting in an M shaped QRS morphology in lead V1 or V2. That means that you will see two upstrokes (R and R') rather than the usual one in these leads. You will also often see a dominant prolonged S wave in V6. In V1 the net direction of the QRS is usually positive (upright) in RBBB (in contrast to it normally being negative), due to the high amplitude R' wave.

Figure 2 below illustrates the diagnostic features of RBBB:

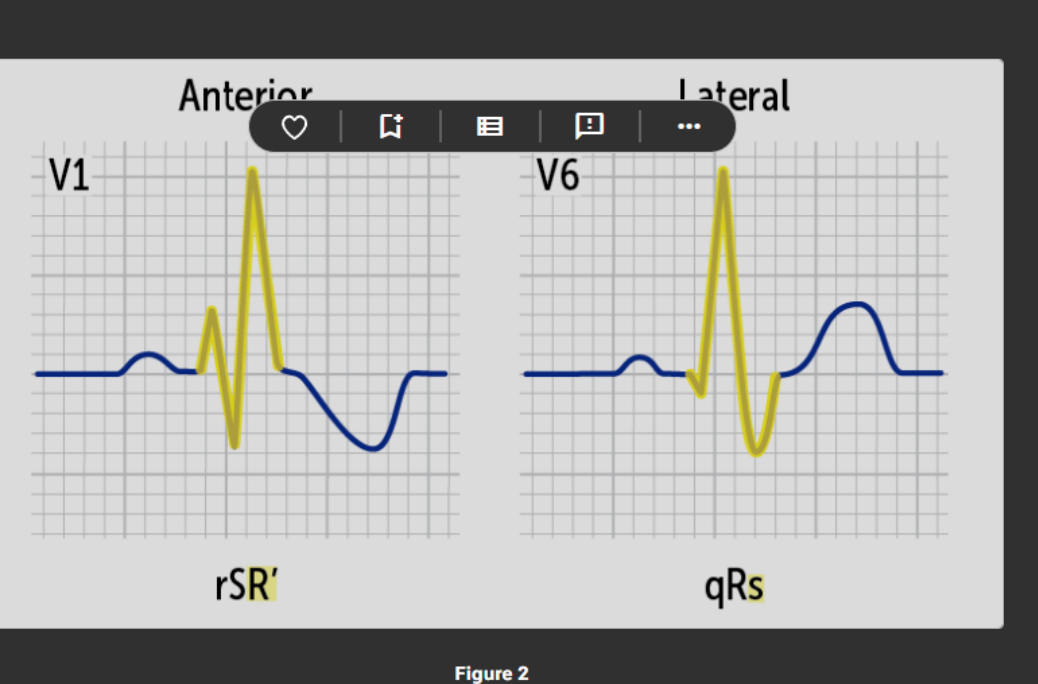


Figure 2

Left bundle branch block (LBBB)

Like RBBB, LBBB causes a widening of the QRS to > 120 ms. Like RBBB, activation of the ventricles is sequential (in *series*) rather than simultaneous. In contrast to RBBB, the overall direction of activation is from right to left, resulting in a downward deflection in lead V1.

Some features of LBBB are shown in figure 3 below:

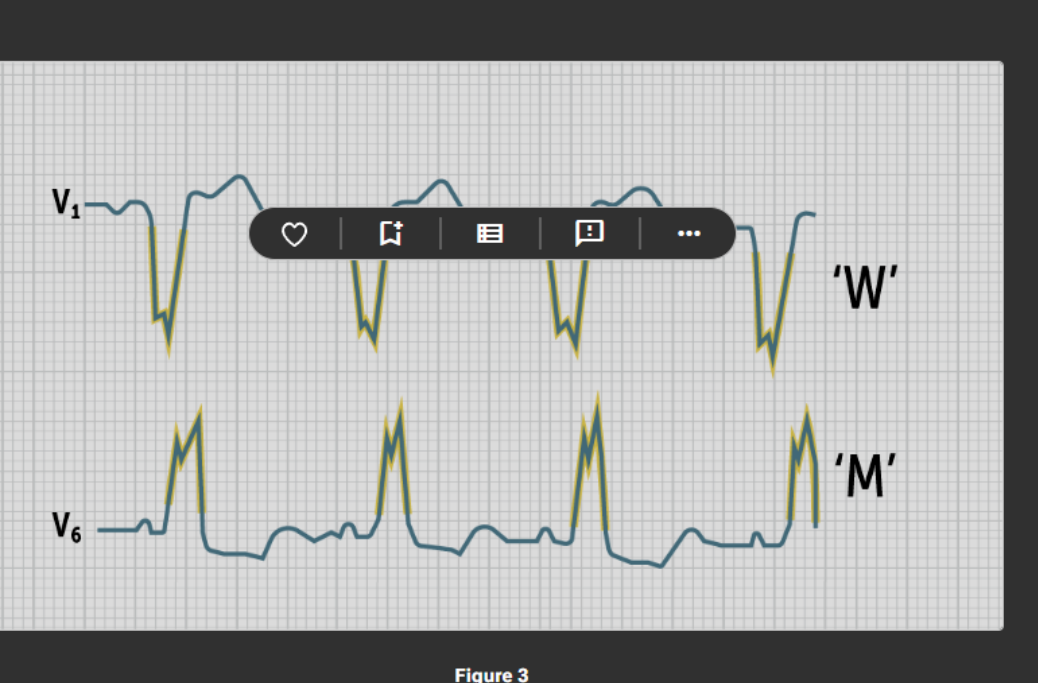


Figure 3

Note the following features in this figure: 1) the QRS is wide; 2) the QRS is downward in V1; 3) notching of the QRS is shown here (W and M patterns) but is not an essential feature of LBBB and is not always present.

REVIEW: to figure out which Bundle Brach Block is causing the wide QRS complex on your patients ECG, focus on lead V1. If there are "Rabbit ears" (rSR'), it's a RBBB. If there is a wide negative deflection in V1 ('W'), it's a LBBB.

Causes

Causes of right and left bundle branch block can include any type of cardiac pathology. In cases where no pathologic process is identified the bundle branch block is considered idiopathic.

Clinical significance and presentation of bundle branch blocks

Although by themsel te symptoms, their clinical significance is three fold: 1) they may be a marker, and in some instances the first sign, of underlying myocardial disease; 2) the dyssynchronous (in *series*) activation of the ventricles can, over time, lead to a reduction in ejection fraction and even heart failure; 3) should the other (contralateral) branch become blocked a very dangerous form of AV block may suddenly result (see discussion below).

Treatment of patients with bundle branch block

Asymptomatic patients with BBB do not require treatment. Patients with heart failure may require resynchronization therapy (simultaneous pacemakers in the RV and LV), and those who progress to second- or third-degree AV block require a pacemaker.

What is an AV block? (LO#1,2, and 4)

AV block is a delay or interruption in conduction through the AV node or through the adjacent Bundle of His. The type of AV block and resultant ECG pattern depends on the degree and site of the block. Remember, the bundle branches are divisions of the Bundle of His. And, even though they extend into the ventricles, they are *functionally*, though not necessarily *anatomically*, considered part of the AV junction. A common point of confusion in discussions of AV block are the multiple ways of classification. Four principal bases for classification are: 1) *degree* of block (first, second and third, discussed immediately below); 2) the *Mobitz type* (1 and 2); 3) the *ratio* of atrial to ventricular impulses (eg 2:1, 3:1); 4) the anatomic level (suprahisian aka AV nodal, intrahisian, ie at the level of the common Bundle of His, and infrahisian meaning bilateral bu ons correlate with one another somewhat. For example, Mobitz type 1 AV block almost always occurs at the anatomic level of the AV node while type 2 usually occurs at lower levels. The Mobitz classification is traditionally only used in cases of second degree block. On the other hand, there is also considerable overlap. For example, third degree AVB can occur at any anatomic level. Different conduction ratios can also be seen at all anatomic levels.

First–Degree Atrioventricular Block

First-degree AV block is defined as a prolonged time between atrial depolarization and ventricular depolarization, leading to a **PR interval of more than 0.2 seconds.**

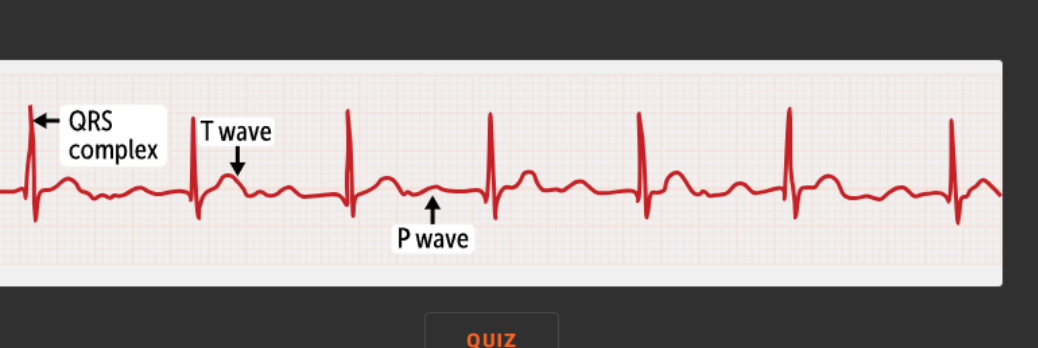
Clinical Presentation. Patients with first-degree AV block are generally asymptomatic. Thus, first degree AVB is strictly an ECG presentation.

Pathophysiology. The electrical signal takes longer to travel down the AV node and Bundle of His. Almost any underlying cardiac disease as well as certain drugs and increased output from the parasympathetic nervous system can cause it.

CLINICAL CORRELATION

Another common cause of first-degree AV block is a physiologic conduction delay in highly conditioned athletes with slow heartbeats (and normal hearts). Normally, when the heart rate slows, the vagal system also slows conduction at the AV node. In these athletes there may be an asymptomatic PR delay longer than 0.2 seconds, which is technically considered a physiologic first-degree AV block. There is no disease or treatmen

Diagnosis. First-degree AV block is diagnosed by a prolonged PR interval of more than 0.2 seconds in adults (figure 4 below).



QUIZ

Figure 4 First-degree atrioventricular block. Note that the PR interval, the interval from the beginning of the P wave to the beginning of the QRS, is >200 ms (one large box).

In first-degree AV block, how long is the PR interval?

Treatment. Most first-degree AV blocks do not require any treatment.

Second–Degree Atrioventricular Block

In second-degree AV block, some (but not all) atrial impulses are unable to reach the ventricles. Second-degree AV Block causes "Grouped Beats" patterns in the rhythm. Second-degree AV block is separated into two types based on the ECG pattern: Mobitz type I (also called Wenckebach) and the more serious Mobitz type II.

Clinical Presentation. Second degree AV block may manifest only as occasional dropped beats and therefore be asymptomatic. However, either form of second degree AV block can progress to high grade or complete heart block. Type II block is the more dangerous of the two in this regard.

Pathophysiology. The two types have different physiology.

Mobitz type I block occurs in the AV node. Causes can be:

- Increased parasympathetic activity (vagal tone)
- Cardiac disease: intrinsic AV nodal disease, myocarditis, inferior MI, and cardiac surgery
- Medications: β-blockers, nondihydropyridine calcium-channel blockers (CCBs), and digoxin

Mobitz type II most commonly occurs **below the AV node** in the Bundle of His or, more often, in the bilateral bundle branches. In addition to the characteristic interm complex will be prolonged if there is a coexisting bundle-branch block.

Mobitz type II is almost always caused by cardiac disease, including MI, cardiomyopathy, and myocarditis.

Diagnosis. Mobitz type I presents with **progressive lengthening of the PR** interval of consecutively conducted beats until a QRS beat is dropped (Figure 5).

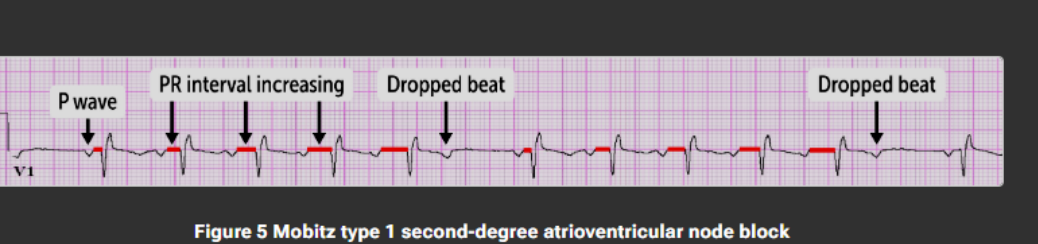
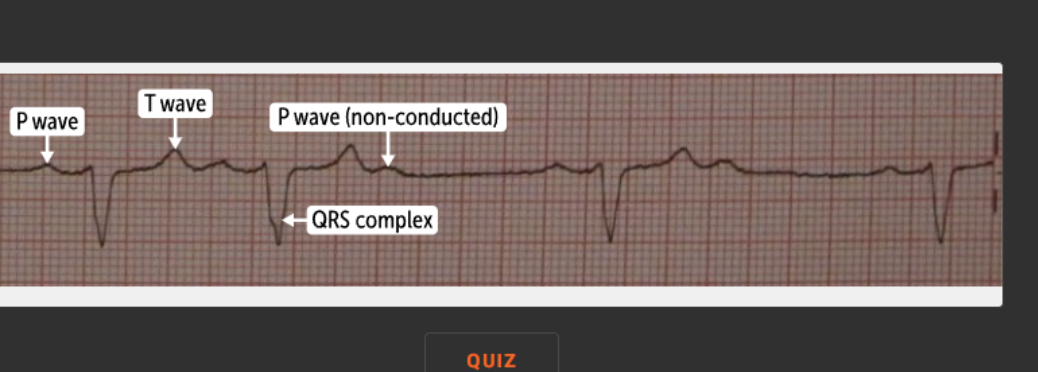


Figure 5 Mobitz type 1 second-degree atrioventricular node block

Mobitz type II, by contrast, presents with constant PR intervals of consecutively conducted beats followed by the dropped beat. (Figure 6 below).



QUIZ

Figure 6 Mobitz type II second-degree atrioventricular node block

Treatment. In asymptomatic Mobitz type I, no treatment is indicated. If the condition is symptomatic due to reduced cardiac output, treatment is needed. Mobitz I, in contrast to type II, tends to occur in acute situations, be dependent on vagal tone or certain drugs, and be reversible, often resolving over time. Atropine, because of its muscarinic blocking (anti-vagal) properties, is usually effective in the short term, serving to buy time until the process resolves. In exceptional cases a pacemaker is needed. In acute situations suspected to resolve a temporary pacemaker will suffice. In the unusual circumstance where type I block is a chronic condition a permanent (implanted) pacemaker is needed.

Mobitz type II, on the other hand, is almost always intrinsic to the heart (not due to temporary external factors) and chronic. It often progresses to third-degree (complete) AV block, and always requires treatment with a **pacemaker** even in asymptomatic patients.

The two types of second-degree AV block are contrasted in Table 1. Remember, "Grouped Beats"

Table 1 Second-degree atrioventricular node block

	Mobitz type I (Wenckebach)	**Mobitz type II**
ECG findings	Progressive lengthening of PR interval of consecutively conducted beats until eventually QRS is dropped	PR interval remains constant in consecutively conducted beats, leading up to the drop.
Anatomic site of block	AV node	Within Bundle of His or bundle hes
Complications	Usually asymptomatic	Heralds more serious conduction abnormalities; may progress to third-degree (complete) AV block

Why does Mobitz type II second-degree AV block have a worse prognosis compared to type I?

Third–Degree Atrioventricular Block

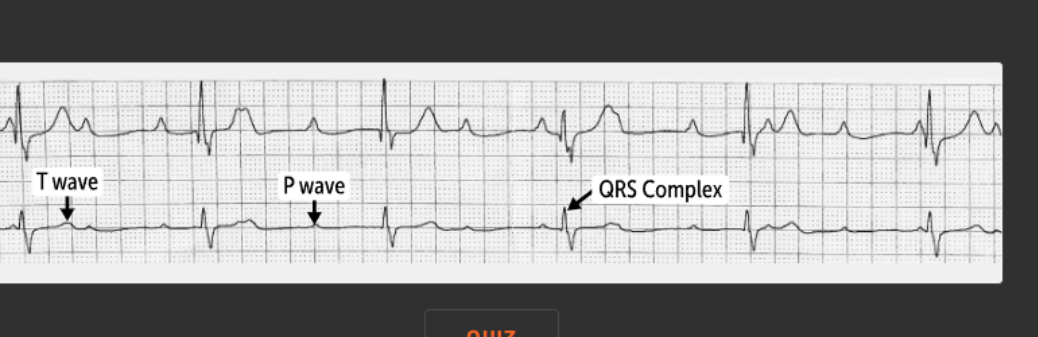
The terms third degree AV block and complete heart block are synonymous. Complete heart block means exactly what it says: there is NO conduction between atria and ventricles. Remember: more P's than QRS's

Clinical Presentation. The symptoms of complete heart block include bradycardia, fatigue, dyspnea, chest pain, syncope, and sudden cardiac arrest, all related to the slow heart rate. Symptoms are generally more severe than in second-degree block.

Pathophysiology. C he AV node, at the Bundle of His, or at infra-Hisian locations. In the absence of a conducted SA node signal, a default "rescue" pacemaker takes over in the ventricles. This abnormal, slow (20-60 bpm) ventricular impulse is known as an **escape rhythm** and can originate at the junction of the AV node/His bundle, within the His-Purkinje system, or from a ventricular myocyte itself. The lower down in the conduction system the block is, the slower will be the available escape pacemaker.

In complete heart block, P waves usually continue but are not conducted and bear no relationship to the QRS complexes. This state of affairs is called *AV dissociation.*

Diagnosis. Third-degree AV block shows P waves that do not correspond with QRS complexes (Figure 7 below).



QUIZ

Figure 7 Third-degree atrioventricular block

That means you can calculate a separate P wave rate and a slower QRS rate.

Blockade at or above w escape QRS complexes on ECG (< a-Hisian locations always produces wide escape QRS complexes on ECG. Remember: more P's than QRS's

Why does the ventricular rate slow in third-degree atrioventricular (AV) block?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to JC, what do you tell him and his parents about his future on the basketball team?

You explain that the ECG showed a first-degree AV block, which means the time between the contraction of the heart's upper chamber (atrium) and lower chamber (ventricle) is prolonged. You go on to say that first-degree AV blocks are generally asymptomatic; the condition is fairly common with highly conditioned, young endurance athletes, so it will not pose a risk to JC playing competitive sports. You do tell JC that if he d ld return to the clinic for evaluation.

Summary

- Heart block is a disorder of the heart's conduction system where impulse conduction is slowed or blocked, often at the AV junction.
- The severity of heart block varies widely, with symptomatic cases more typically affecting elderly patients or those with preexisting cardiovascular conditions.
- Heart block can be in the form of exit block, bundle branch block, or AV block.
- First-degree AV block is characterized by a prolonged PR interval (>0.2 seconds) and is often asymptomatic, requiring no treatment.
- Second-degree AV block includes Mobitz type I and II, with Mobitz type II requiring pacemaker treatment due to risk of progression to complete heart block.
- Third-degree AV block results in complete heart block with slow ventricular escape rhythm, necessitating pacemaker placement.

Review Questions

1. Which of the below best characterizes first-degree atrioventricular block?

A. ○ Dropped P waves
B. ○ Narrow Q
C. ● PR interval of >0.2 seconds
D. ○ ST segment of >0.2 seconds
E. ○ Widened QRS complex

Hide Explanation

The correct answer is PR interval of >0.2 seconds (C). First-degree atrioventricular (AV) block is defined as a PR interval of >0.2 seconds. Dropped P waves (A) appear in second-degree AV block. Narrow QRS complex (B) means that the cardiac conduction began at or above the Bundle of His. ST segment of more than 0.2 seconds (D) is not used in the diagnosis of first-degree AV block. Widened QRS complex (E) means that the cardiac conduction began below the Bundle of His.

2. All but which of the following are true of bundle branch blocks:

A. ○ Left bundle branch block has a downward QRS in lead V1.
B. ○ Right bundle branch block displays a prominent and wide S wave in lateral leads due to delayed activation of the right ventricle.
C. ● Left bundle branch blocks have prominent Q waves.
D. ○ Both right and left bundle branch blocks have a wide QRS (≥ .12 sec, or 120 ms).

Hide Explanation

The correct answer is Left bundle branch blocks have prominent Q waves (C). In fact, the "physiologic" Q waves seen in the normal electrocardiogram, which reflect initial activation of the septum, are *lost* in LBBB. Such early septal activation is dependent on rapid conduction through the left bundle. The other choices, (A), (B), and (D), all reflect morphologic properties of bundle branch blocks cited in the text above.

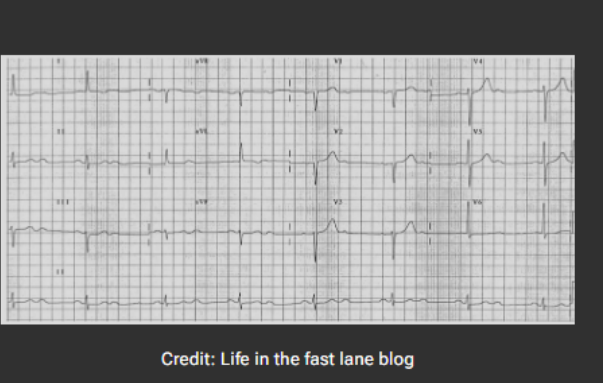
3. A patient presents to the emergency department for evaluation of a syncopal episode. He describes a sudden blackout from which he recovered after a few seconds of lying down. His heart rate is 40 beats/min. The ECG shows normal P wave morphology. The P wave rate is 90/minute, while the QRS rate is 40 beats/min. What is a feature of this type of heart block that is not seen with any of the other types of heart block?

A. ● Atrioventricular dissociation
B. ○ Dropped P waves
C. ○ Episodic supraventricular tachycardia
D. ○ Prolonged PR interval

Hide Explanation

The correct answer is inct, unrelated P and QRS rates suggest co G finding is known as AV dissociation, as is present in all forms of complete AV block. (AV dissociation, though always present in complete AV block, is not unique to complete AV block, also often seen in other arrhythmias such as VT, AIVR, some junctional rhythms and some others. Discussion of all the types of AV dissociation is beyond the scope of this brick). Dropped P waves (B) are seen in Mobitz type I and type II second-degree heart block. Heart block does not lead to episodic supraventricular tachycardia (C). Prolonged PR interval (D) characterizes first-degree heart block.

4. A 60 year old man presented to the ER with dizziness and the electrocardiogram shown below. Blood chemistries, ABGs, CBC, and chest xray were normal. He was on no medication. Which of the following is true concerning this patient?



Credit: Life in the fast lane blog

A. ● We have enough information to recommend permanent pacing.
B. ○ Since the PR intervals are constant, this is Mobitz II block
C. ○ This is co
D. ○ The patient should undergo MRI of the brain and internal auditory canals to evaluate for a neurologic or inner ear cause of dizziness.

Hide Explanation

The correct answer is A. Due to the fact that this is a fixed condition (no drugs or electrolyte condition, or acute MI to account for it), is symptomatic (dizziness), and has a dangerously slow ventricular rate (43), permanent pacing is indicated. Option B is incorrect, but it is a common mistake. Although it is true that the PR intervals are constant, this cannot be applied as a criterion for the Mobitz classification. In order to apply the Mobitz classification here, one must see a constant PR in *consecutively conducted beats* leading up to the drop, which we do not see here. (Because this block is 2:1, only every other beat is conducted. There are no *consecutively* conducted beats). The Mobitz criteria cannot be applied, and this could be either type I or II. Option C is incorrect. The fact that there is a relationship between the P wave and the QRS of every other beat, AV dissociation, and hence complete heart block, is not present. Option D is incorrect. There is no clinical evidence in the history of vertebro-basalar or vestibular disease. Moreover, the slow heart rate is plausible enough evidence that AV block is the culprit.

5. Disease processes in the heart that could result in heart block include which of the following?

A. ○ Ischemia
B. ○ Scar tissue
C. ○ Infiltrative
D. ○ Inflammatory processes (eg Lyme disease)
E. ● All of the above

Hide Explanation

All are correct, reflecting the fact that any pathologic process in the heart is capable of impairing conduction, thus producing block.

How would you rate this Brick?

References

- Harkness WT, Hicks M. Right bundle branch block. In: *StatPearls*. StatPearls Publishing. Updated August 13, 2021. Accessed September 20, 2022. https://www.ncbi.nlm.nih.gov/books/NBK507872/ https://www.uptodate.com/contents/sinoatrial-nodal-pause-arrest-and-exit-block#H1083507
- Scherbak D, Shams September 20, 2022. https://www.ncbi.nlm.nih.gov/books/NBK507872/ https://www.uptodate.com/contents/sinoatrial-nodal-pause-arrest-and-exit-block#H1083507
- Scherbak D, Shams P, Hicks GJ. Left Bundle Branch Block. [Updated 2024 Oct 5]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK482167/
- **Arrhythmia Essentials, Ch 2, available via Clinical Key**

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Compare and contrast the clinical presentation and pathophysiology of the different types of heart blocks~~
02 ☑ ~~Differentiate the ECG patterns associated with 1st, 2nd, and 3rd degree heart blocks~~
03 ☑ ~~Differentiate the ECG patterns associated with bundle-branch blocks~~
04 ☑ ~~Outline treatment plan for heart blocks and bundle branch blocks.~~

Objective Confidence: 4/4